Natalie M. Allen

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EDUCATION

Purdue University, West Lafayette, Indiana USA — January 2023 – Present (expected July 2027)
Ph.D. Candidate in Evolutionary Biology and Conservation Genomics
Dissertation: *Genomic approaches to investigating evolutionary response: insights into diversity, demography, and divergence*

Marquette University, Milwaukee, Wisconsin USA — August 2019 – December 2022
B.S. in Biological Sciences with Comprehensive Honors, *magna cum laude*
Minor in Environmental Studies

RESEARCH EXPERIENCE

Department of Forestry and Natural Resources, Purdue University — West Lafayette, Indiana USA
Ph.D. Candidate, DeWoody Lab — January 2023 – Present

- Lead investigator on whole-genome resequencing projects that apply population genomics to the conservation and management of at-risk vertebrates.
- Led a population genomic study of the western chicken turtle, identifying cryptic population structure relevant to conservation unit designation; shared results directly with Texas Parks & Wildlife Department, which supported the work in part because of the species' active ESA status review.
- Led a genomic analysis of Pacific gray whale population structure, detecting ancestry from a potential ghost population; presented findings to the International Whaling Commission's Scientific Committee.
- Studying the conservation genomics of Montezuma quail in collaboration with a U.S. Fish & Wildlife Service biologist; helped develop a landscape-level pangenomic framework linking structural-variant adaptation to climate-driven vulnerability (PNAS, *in revision*) and am leading a dissertation chapter on range-wide genomic erosion.
- Co-developed frameworks for incorporating genomic data into conservation monitoring and formal threat assessments, and evaluated short-read pangenomes for conservation applications.
- Assembled a novel vertebrate reference genome and long-read pangenomes; developed reproducible pipelines for variant calling, demographic inference, and structural variant analysis.
- Mentor undergraduate researchers in genomic analysis and scientific communication.

Department of Biological Sciences, Marquette University — Milwaukee, Wisconsin USA
Undergraduate Researcher, Gamble Evolutionary Biology Lab — September 2019 – December 2022

- Led an independent project to study the evolution of visual opsin genes across geckos using phylogenetic trees, selection tests, and molecular clock analysis (August 2020 – December 2022).
- Completed DNA extraction, PCR, gel electrophoresis, multi-sequence alignment, and phylogenetic analysis to identify hybrid geckos (September 2019 – March 2020).

SKILLS

- **Bioinformatics:** Pre-processing, variant calling, and population genomic analysis of whole genome resequencing data; genome assembly; phylogenomics and comparative genomics; pangenomics using PGGB and Minigraph-Cactus; Nextflow pipeline construction; R; Python; shell scripting; ArcGIS Pro; R and Python for GIS.
- **Laboratory Genetics:** High molecular weight DNA extraction; cDNA synthesis; PCR; gel electrophoresis; DNA cloning; RT-qPCR; quality control (Qubit, TapeStation).

- **Field Work:** Field identification of North American vertebrates (especially herpetofauna); freshwater turtle trapping and tissue sampling.
- **Climate & Spatial Analysis:** Integration of genomic and environmental datasets; time-series modeling; SLiM simulations; spatial visualization.
- **Leadership & Communication:** Undergraduate mentorship; graduate teaching including lecturing (statistics for natural resources); science communication training; peer review for international journals; public writing.

SCIENTIFIC PUBLICATIONS

Andrew N. Black, Jong Yoon Jeon, Andrew J. Mularo, **Natalie M. Allen**, Erangi Heenkenda, Julia C. Buchanan-Schwanke, John W. Bickham, Zachary E. Lowe, J. Andrew DeWoody. (2025) Thematic layers of genomic susceptibility for conservation monitoring. Molecular Ecology 34(23), e17582. https://doi.org/10.1111/mec.17582

Jong Yoon Jeon, **Natalie M. Allen**, Andrew N. Black, J. Andrew DeWoody. (2026) Short-Read Pangenomes and Their Potential Utility in Population and Conservation Genomics. Molecular Ecology Resources e70060. https://doi.org/10.1111/1755-0998.70060

Natalie M. Allen, Andrew J. Mularo, Jong Yoon Jeon, Alvaro G. Hernandez, Ethan S. Streckfuss, Wade A. Ryberg, Toby J. Hibbitts, Brandon C. Bowers, Paul Crump, and J. Andrew DeWoody. (2026) Whole genome resequencing data provide high-resolution conservation insights into population structure and life history. Conservation Genetics 27, 61. https://doi.org/10.1007/s10592-026-01777-8

Jong Yoon Jeon, Andrew J. Mularo, **Natalie M. Allen**, Erangi Heenkenda, John W. Bickham, Andrew N. Black, J. Andrew DeWoody. (2026) Genomic metrics should inform formal conservation threat assessments. Trends in Ecology and Evolution. https://doi.org/10.1016/j.tree.2026.06.015

Natalie M. Allen, Anna Brüniche-Olsen, Jong Yoon Jeon, Andrew J. Mularo, John W. Bickham, Jorge Urbán Ramírez, J. Andrew DeWoody. (2026) The contemporary Pacific gray whale (Eschrichtius robustus) gene pool includes ancestry from a potential ghost population: inferences from population genomics. Journal of Heredity. Advance online publication https://doi.org/10.1093/jhered/esag059

Erangi Heenkenda, Andrew N. Black, Craig Stockwell, **Natalie M. Allen**, J. Andrew DeWoody. (2026) Temporal population genomics reveals evidence of purging genetic load in small, isolated populations of conservation concern. Proceedings of the Royal Society B. *In revision.*

Jong Yoon Jeon, **Natalie M. Allen**, Audrey J. Heckel, Alberto Macias-Duarte, René Jesús Toscano-Meléndrez, Jesús Raúl Martínez-Sandoval, Angel Montoya, John McLaughlin, J. Andrew DeWoody. (2026) A landscape level pangenomic framework reveals adaptation based on structural variants and climate-driven vulnerability in a sedentary bird. PNAS. *In revision.* Preprint: https://ecoevorxiv.org/repository/view/12415/

Natalie M. Allen, Mekala Sundaram, Janna Willoughby, Avril Harder, Jong Yoon Jeon, J. Andrew DeWoody. (2026) Genomic divergence between species identifies taxonomic outliers and establishes calibrated molecular clocks. PNAS. *In review.* Preprint: https://doi.org/10.32942/X2T11S

CONFERENCE AND SEMINAR TALKS

- The contemporary Pacific gray whale (Eschrichtius robustus) gene pool includes ancestry from a potential ghost population: inferences from population genomics. *Evolution* 2026, Cleveland, Ohio, USA.
- The contemporary Pacific gray whale (Eschrichtius robustus) gene pool includes ancestry from a potential ghost population: inferences from population genomics. *International Whaling Commission Scientific Committee* 2026, Bled,

Slovenia.
- Whole genome resequencing data provide high-resolution conservation insights into population structure and life history. *Evolution* 2025, Athens, Georgia, USA.
- Conservation insights from population genomics in an elusive turtle. Department of Biological Sciences *Ecolunch* 2025, Purdue University.
- Population genomics of the Pacific gray whale. *Evolution* 2024, Montreal, Canada.
- Population genomics of the gray whale. *Midwest Population Genetics* 2023, University of Michigan.

POSTER PRESENTATIONS

- Population genomics can inform conservation efforts in an elusive turtle. Midwest Population Genetics 2024, Indiana University.
- Conservation genomics of the western chicken turtle. Department of Forestry and Natural Resources 2023 poster symposium, Purdue University.
- The evolution of gecko vision. Department of Biological Sciences 2022 Summer Research Symposium, Marquette University.
- The evolution of gecko vision. Evolution 2022, Cleveland, Ohio, USA.
- Evidence of molecular convergence and positive selection on visual opsin proteins in geckos. Department of Biological Sciences 2022 Undergraduate Research Day, Marquette University.
- Testing for molecular convergence and selection on gecko opsin genes. Honors Undergraduate Research Fair 2021, Marquette University.
- The evolution of gecko vision: Molecular convergence in visual opsin proteins. Department of Biological Sciences 2021 Undergraduate Research Day, Marquette University.

FELLOWSHIPS AND AWARDS

- Kirkpatrick Memorial Graduate Student Award, 2026 $3,000
- Welder Wildlife Foundation Research Fellowship, 2026–27 $45,000
- D. Woods Thomas Memorial International Support Fund, 2025 $1,500
- Purdue Institute for a Sustainable Future Travel Award, 2024 $650
- Marquette Biological Sciences Conference Travel Award, 2022 $1,500
- Marquette Biological Sciences Summer Research Program, 2022 $4,000
- Marquette Honors Summer Research Fellowship, 2021 $3,500

WORK AND TEACHING

Purdue University

Graduate Research Fellow, Welder Wildlife Foundation — January 2026 – Present
Graduate Teaching Assistant, Statistics for Natural Resources — Spring 2026
Graduate Research Assistant, DeWoody Lab — January 2023 – December 2025

Marquette University

Teaching Assistant, Principles of Biological Investigation — Fall 2022
Teaching Assistant, Evolutionary Biology — Spring 2022
Teaching Assistant, Cellular Biology — Fall 2021
Squamate Animal Caretaker, Gamble Lab — Aug. 2020 – Dec. 2022
- *Manager* (May 2022 – Dec. 2022)

Varsity Tutors

Tutor, ACT, SAT, Algebra, Precalculus, Calculus 1 — July 2019 – Dec. 2022

Independent Tutoring
Tutor, English for ESL students, Algebra Feb. 2016 – Dec. 2022

UNDERGRADUATE MENTEES

- Ethan Streckfuss August 2024 – May 2025
  - Poster presentation: Genomic insights into the conservation of the western chicken turtle, Spring Undergraduate Research Expo 2025, Purdue University, USA
- Brady Kahner (co-mentored) August 2024 – Present
- Megan Johnson January 2023 – May 2024
  - Poster presentation: Nuclear copies of Mitochondrial DNA fragments (NUMTs) in the Gray Whale Genome, Fall Undergraduate Research Expo 2023, Purdue University, USA
  - Poster presentation: Nuclear copies of Mitochondrial DNA fragments (NUMTs) in the Gray Whale Genome, Society for Molecular Biology and Evolution "Molecular Evolution in Small Populations" 2023, Princeton University, USA
- Rafaela Besse January 2023 – December 2023
  - Poster presentation: Nuclear copies of Mitochondrial DNA fragments (NUMTs) in the Gray Whale Genome, Fall Undergraduate Research Expo 2023, Purdue University, USA
  - Poster presentation: Nuclear copies of Mitochondrial DNA fragments (NUMTs) in the Gray Whale Genome, Society for Molecular Biology and Evolution "Molecular Evolution in Small Populations" 2023, Princeton University, USA

ATTENDED TRAININGS

- Workshop: Communicating Science to Diverse Audiences, Evolution Meeting June 2025
- Workshop: GIS and Data Science, Purdue University October 2024
- Workshop: Designing Nextflow pipelines, Purdue University January – May 2024
- Course: Analysis of high throughput data for population genomics, University of Copenhagen (virtual) August 2023
- Workshop: SLiM forward evolutionary simulation software, Albuquerque, New Mexico June 2023
- Training: RT-qPCR, Purdue FNR Genetics Lab May 2023

SCIENCE COMMUNICATION AND MEDIA

- Purdue Department of Forestry and Natural Resources Graduate Student Spotlight (2026). Graduate Student Spotlight
- **Allen, N.** (2020, July 30). *Allen: MU must offer remote learning, teaching options for Fall Semester.* Marquette Wire. Article link

PEER REVIEW CONTRIBUTIONS

- Science (co-reviewer)
- Molecular Ecology
- Molecular Ecology Resources

SERVICE AND LEADERSHIP

- *College of Agriculture Representative*, FNR Graduate Student Council May 2025 – Present

- *Activity organizer*, Purdue 2026 Grandparents University 2026
- *Judge*, Fall Undergraduate Research Expo 2024
- *Organizer*, Lab Bioinformatics Coffee Hours Jan. 2024 – Present
- *Volunteer*, Wisconsin Humane Society 2022
- *Founder*, Biology Research Achievement Mentorship Program (RAMP) 2022
- *Interviewer*, Building DEI in Marquette Honors Admissions 2022
- *Vice President*, Marquette Waterski Team 2022
- *Women's Captain*, Marquette Waterski Team 2021
- *Volunteer*, River Revitalization Foundation 2020 – 2021
- *Volunteer*, Milwaukee Public Museum 2019 – 2020